

The expanded $|\mathfrak{B}_2|^2$ bracket

All 24 transverse terms and the four colour terms, checked

$$A = (x - z)^2, B = (x - w)^2, C = (z - w)^2, A' = (x' - z)^2, B' = (x' - w')^2, C' = (z - w')^2, S = p^+ + k^+$$

Yes — the expansion is correct. Every one of the six longitudinal groups reproduces $\mathcal{W}^{ji}(x, z, w) \mathcal{W}^{ji}(x', z, w')$ exactly: agreement to 10^{-14} – 10^{-16} on three random configurations with independent p^+, k^+ (§1). All 24 transverse terms, every denominator and every sign, are right. The colour bracket is right too, and **your N_c term is exactly what the identity gives** (§2). Three items remain, one of them a correction to what I gave you (§3).

1. The transverse bracket, group by group

Writing the vertex as $\mathcal{W}^{ji} = \mathcal{T}_1 + \frac{S}{k^+} \mathcal{T}_2 + \frac{S}{p^+} \mathcal{T}_3$, your six groups are exactly the six contractions:

your group	is	trial 1	trial 2	trial 3	max diff
$d_\perp$	$\mathcal{T}_1 \cdot \mathcal{T}'_1$	−25.912419143	6.187400097	−0.478337090	$4 \cdot 10^{-15}$
S/k^+	$\mathcal{T}_1 \cdot \mathcal{T}'_2 + \mathcal{T}_2 \cdot \mathcal{T}'_1$	67.617138672	−36.878229974	−0.371158127	$4 \cdot 10^{-16}$
S/p^+	$\mathcal{T}_1 \cdot \mathcal{T}'_3 + \mathcal{T}_3 \cdot \mathcal{T}'_1$	30.159517121	−4.867642185	1.730598948	$7 \cdot 10^{-15}$
S^2/k^{+2}	$\mathcal{T}_2 \cdot \mathcal{T}'_2$	−14.691384030	57.811047161	20.678928345	$4 \cdot 10^{-15}$
S^2/p^{+2}	$\mathcal{T}_3 \cdot \mathcal{T}'_3$	−24.539341717	1.190220481	0.058236503	$2 \cdot 10^{-14}$
$S^2/p^+ k^+$	$\mathcal{T}_2 \cdot \mathcal{T}'_3 + \mathcal{T}_3 \cdot \mathcal{T}'_2$	−143.375770329	15.097295787	−0.387351441	$3 \cdot 10^{-14}$
total	$\mathcal{W}^{ji} \mathcal{W}'^{ji}$	−110.742259425	38.540091367	21.230917138	$3 \cdot 10^{-14}$

Each group matches *individually*, not merely in the sum — so no error has cancelled between groups. A few structural checks worth keeping in mind, all of which your expansion respects:

- The $S^2/p^+ k^+$ group has **eight** terms, not four, because $\mathcal{T}_2 \mathcal{T}'_3$ and $\mathcal{T}_3 \mathcal{T}'_2$ are different objects. You have all eight, with the right relative signs: $(+, -, +, -)$ for $\mathcal{T}_3 \mathcal{T}'_2$ and $(+, +, -, -)$ for $\mathcal{T}_2 \mathcal{T}'_3$.
- Denominators: the $\frac{1}{4}$ terms always pair the two "self-energy-like" factors (BB' , AA' , AB' , $A'B$), the $\frac{1}{2}$ terms always pair one of those with a C or C' , and the leading terms are always CC' . Every one of yours follows this.
- The S/k^+ group's $\frac{1}{4}$ terms carry B and B' ; the S/p^+ group's carry A and A' — and the signs flip between the two groups ($-$ for S/k^+ , $+$ for S/p^+), which is the sign in $\mathcal{V}_w$ vs $\mathcal{V}_z$. Correct in both.

On $d_\perp$. Keeping it symbolic is harmless, but since this row has no UV pole there is nothing for an $O(\epsilon)$ piece to hit: $d_\perp = 2$ exactly. You can set it now.

2. The colour bracket — and your N_c term

All four combinations are present and correctly assembled. The fourth is the one you collapsed:

$$\sum_b U^{bc'}(z) U^{bc}(z) = \delta^{c'c} \implies \sum_{a,c'} f^{ac'd} f^{ac'd} = N_c \delta^{d'd} \implies N_c U^{e'd}(x') U^{de}(x)$$

Verified to $4 \cdot 10^{-15}$: **given the bra ordering you wrote, your term 4 is exactly right.** It works because both Wilson lines already sit at the *same* point z and share the summed index b — which is why it needed no limit; it holds for all w, w' .

But it inherits the transpose. Your bra carries $U^{e'd}(x')$ (ρ index first) while your ket carries $U^{de}(x)$ (f index first). Conjugating the ket does not transpose a real orthogonal matrix, so the bra should read $U^{d'e'}(x')$. With that ordering the same collapse gives

$$C_{22}^{e'e} = N_c [U^\dagger(x') U(x)]^{e'e}$$

(verified to $1 \cdot 10^{-14}$) — the adjoint **dipole**, which is what you want physically. The two differ by $\|N_c U(x') U(x) - N_c U^\dagger(x') U(x)\| = 3.72$, so it is not cosmetic. The same transpose sits in your term 3.

3. Three remaining items

1. **The measure** is still $\int_{z,w,w'}$, but x and x' appear throughout — in every kernel and in $\rho^d(x)$, $\rho^{e'}(x')$. It must be $\int_{x,x',z,w,w'}$.
2. **My own prefactor slip, which you should fix in the result I sent.** Your row carries $\int \frac{dp^+}{2\pi}$, and the boxed answer in the previous note folded the p^+ integral into $\mathcal{M}_{ab}$ without that $\frac{1}{2\pi}$. The overall constant there should read

$$\frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \cdot \frac{1}{2\pi} = \frac{g^4}{64\pi^8} \quad \text{not} \quad \frac{g^4}{32\pi^7},$$

a factor 2π . Everything inside — the six masters, their closed forms, the $\frac{1}{4k^+} \log \frac{\sqrt{-k^+}}{\Lambda}$ transverse log — is unaffected.

3. **The $(2\rho) \times (2\rho)$ piece of $|\mathfrak{B}_2|^2$ is still missing.** What you have written here is the complete $(1\rho) \times (1\rho)$ square — all four colour terms — so this expression is internally complete. But $|\mathfrak{B}_2|^2$ also has the $(1\rho) \times (2\rho)$ cross terms (your N_1) and the $(2\rho) \times (2\rho)$ square, coefficient $\frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{p^+k^+}$. The last one matters for the $|z| \rightarrow \infty$ log, since it carries the same $1/|z|^2$ tail.

4. Where this leaves the row

Nothing in the expansion changes the conclusions of the previous note, because those were derived from $\mathcal{W}^{ji}\mathcal{W}'^{ji}$ itself, which is what your bracket equals. To restate them against the expanded form:

- **No transverse UV** in any of the 26 collapsing regions.
- **One transverse IR log**, at $|z| \rightarrow \infty$, carried by the S^2/k^{+2} group at fixed p^+ and by the S^2/p^{+2} group after the p^+ integration; the $\log^2$ between them cancels, leaving $\frac{1}{4k^+} \log \frac{\sqrt{-k^+}}{\Lambda} \frac{(x-w) \cdot (x'-w')}{(x-w)^2 (x'-w')^2} \frac{1}{|z|^2}$.
- **No DGLAP**: the ζ weight of that log is $1/(\zeta\bar{\zeta})$, not $C_{UV}(\zeta) = P_{gg}/2N_c$.

Checked by `src/row18_expand.py`; output in `RESULTS_row18.txt`.